

Animal Implicatures: A Note on the Informativity Principle and Bayesianism*

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Abstract

Several recent analyses of animal signals have proposed that when animals have the choice between a signal S and a more informative signal S^+ , S^+ is preferred if it is true (= Informativity Principle). The principle is usually taken to apply both in production and in comprehension. But some uses of the principle in comprehension can be replaced with simple Bayesian principles. We explain in which cases this happens, and why the key data adduced for the Informativity Principle cannot be explained in this way because they involve production rather than comprehension. In humans, the Informativity Principle in production derives from the speaker’s reasoning on the addressee’s Bayesian behavior, but this derivation lacks plausibility in animals that lack a theory of mind. This suggests a new division of labor in which the animal Informativity Principle is primitive in production, but derived from Bayesian reasoning in comprehension.

Keywords: animal linguistics, animal communication, animal semantics, animal signals, Bayesian pragmatics, Informativity Principle

Statement on the use of AI. For the present version, AI (notably, Claude) was solely used to provide feedback and critical remarks. We experimented with more radical uses of AI, for instance to perform simulations and write a separate text. We ended up not making use of these results.

1 The Informativity Principle in animal linguistics

Several recent analyses of animal signals developed within “animal linguistics” (e.g. Schlenker et al. 2014, 2016a,b,c) have postulated an Informativity Principle:

(1) **Informativity Principle** (from Schlenker et al. 2016c)

Assume that the semantics yields a relation ‘is strictly more informative than’ on some signals that are alternatives to each other. Underinformative signals are prohibited by the following rules:

- a. *Production:* Do not utter S in a situation w if a strictly more informative alternative S^+ is true in w .
- b. *Comprehension:* If you perceive S in a situation w , infer that every strictly more informative alternative S^+ is false in w .

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In its human incarnation, the Informativity Principle derives “implicatures”: an expression S that competes with a more informative expression S^+ is pragmatically enriched to S and not S^+ (e.g. Grice 1989; Horn 1972; Gazdar 1979; Hirschberg 1991; Sauerland 2004; Schulz and van Rooij 2006; Spector 2007; Frank and Goodman 2012; Goodman and Stuhlmüller 2013; Franke and Jäger 2016; Degen 2023, among many others).

A key motivation for the animal version of the Informativity Principle comes from the stylized observation in (2a).

(2) **Stylized motivation for the Informativity Principle**

Several species have a call S used for diverse dangerous situations (e.g. situations where there is a predation and non-predation-related situations), and a call S^+ used only for specific predators, typically eagles. Then:

- a. *Production* [actual argument]: When there is an eagle stimulus, signalers standardly produce sequences of S^+ signals, not of S signals.
- b. *Comprehension* [possible argument]: When there is an S signal, receivers behave as if there were a non-eagle-related threat.

There is considerable evidence that animals perform Bayesian reasoning, which is essential to fitness (see for instance Valone 2024 for a recent review). It is thus natural to ask whether Bayesian principles could obviate the need for the Informativity Principle. We will make two key points.

1. The Informativity Principle in comprehension (= (1b)) can be replaced by Bayesian reasoning. Specifically, even in the absence of an informativity principle either in production or in comprehension, Bayesian reasoning in comprehension may be enough to explain why an S-signal tends to be associated with non-eagle related threats – a point already made in Franke (2009, 2011) in the case of human pragmatics. If the data motivating the Informativity Principle in animals were uniformly based on comprehension, the principle could thus be eliminated. But inspection of the key animal data shows that this is not so. As a result:
2. The Informativity Principle in production (= (1a)) must remain as primitive. In human language, game-theoretic frameworks such as the Rational Speech Act model (RSA) derive the Informativity Principle in production from the speaker’s reasoning on the addressee’s Bayesian behavior. In animals that lack a theory of mind, this derivation is implausible.

Our dual result can motivate a new division of labor: if recent arguments for signal competition in animals are correct, the Informativity Principle in production is primitive, while in comprehension it can be derived from Bayesian reasoning.

Two disclaimers before we start. First, it is not our goal to relitigate the literature on animal linguistics; we will only summarize the key arguments (keeping the call names used in the original studies for ease of cross-reference). There are detailed formal analyses in the literature, carefully weighing the pros and cons of the Informativity Principle. Our argument is conditional: *if* these analyses are on the right track, the Informativity Principle in comprehension can be obviated, but the Informativity Principle in production cannot be.

Second, for an argument for the Informativity Principle as stated in (2) to get off the ground, one signal must be strictly more informative than another signal. But it is hard to find incontrovertible evidence for this situation. For instance, in Titi monkeys, the A-call has been analyzed as raising a serious non-ground threat, while the B-call has been taken to be used for any threat (Schlenker et al. 2016b). The effect of the Informativity Principle could be mimicked without it by just positing that the B-call means: non-[serious non-ground] threat. This would trivialize the entire discussion that follows, as call meanings without additional principles would suffice to derive the observed behaviors. One can take this alternative B-meaning to be unnatural, but it is hard to rule it out. Furthermore, Steinert-Threlkeld et al. 2021 show that under certain evolutionary assumptions, general calls might end up acquiring a specialized meaning of precisely this kind. To ensure that general calls are stable despite this result, one could posit that some of the undesirable specialized meanings are cognitively impossible, and/or that situations of uncertainty regarding the precise nature of the alert make general calls essential. We have nothing to contribute to this discussion in the present piece, and in the remainder we assume that general calls are real.

2 A simple Bayesian alternative in comprehension

Let us consider again a Titi monkey that hears a B-call, and let us assume that there is no Informativity Principle whatsoever, be it in production or in comprehension. Let us further assume that a signaling Titi has the following behavior:

- If there is an eagle, use A with probability .5 and B with probability .5
- If there is a non-eagle-related threat, use B with probability 1.

Now suppose that relative to all threats, eagle-related threats occur with probability .5, while non-eagle-related threats occur with probability .5. These assumptions are summarized in Table 1.

Situations	Eagle ($p = .5$)	Non-eagle ($p = .5$)
Calls	A ($p = .25$) B ($p = .25$)	B ($p = .5$)

Table 1

It is clear that a Titi receiver that hears a B-call will take it to be more likely that there is a non-eagle rather than eagle threat, since conditional on hearing B, the probability of encountering a non-eagle is twice that of encountering an eagle, as shown in Table 1 (the probability of ‘Eagle & B’ is 0.25, while that of ‘Non-eagle & B’ is 0.5). More precisely, by inspecting Table 1, one sees that $P(\text{Non-eagle}|B) = \frac{P(\text{Non-eagle} \& B)}{P(B)} = \frac{0.50}{0.75} = 2/3$. But this means that the effect of the Informativity Principle in perception (= (1b)) is emulated without any Informativity Principle.¹

It is clear that this conclusion does not necessarily follow. If non-eagle threats have probability .1 rather than probability .5, as illustrated in Table 2, while eagle-related threats give rise to half of A-behavior and half of B-behavior, as before, the reasonable reaction when hearing a B-call is to think that there is probably an eagle, not that there is a non-eagle, since we now have $P(\text{Non-eagle}|B) \simeq 0.18$ (cf. Table 2: $P(\text{Non-eagle}|B) = \frac{P(\text{Non-eagle} \& B)}{P(B)} = \frac{0.1}{0.55} \approx 0.18$)

Situations	Eagle ($p = .9$)	Non-eagle ($p = .1$)
Calls	A ($p = .45$) B ($p = .45$)	B ($p = .1$)

Table 2

Let us now consider things in a slightly more general case, as described in (3).

(3) General case

a. Situations

Among all threats, eagle threats occur with probability e , non-eagle threats with probability $(1 - e)$, with $0 \leq e < 1$:

$$P(\text{Eagle}) = e, \quad P(\text{Non-eagle}) = 1 - e$$

b. Calling strategies

Eagle threat situation: use A with probability $1 - b$, B with probability b , with $0 < b < 1$:

$$P(A | \text{Eagle}) = 1 - b; \quad P(B | \text{Eagle}) = b$$

Non-eagle threat situation: use B with probability 1

$$P(A | \text{Non-eagle}) = 0; \quad P(B | \text{Non-eagle}) = 1$$

¹In Franke’s Iterated Best Response model, this derivation corresponds to a level-1 listener who responds to a level-0 (literal) speaker (Franke, 2009, 2011).

Now we can use Bayes's rule to compute the conditional probability of being faced with an eagle vs. a non-eagle threat contingent on hearing A or B.

(4) **Bayes's rule**

$$P(X | Y) = \frac{P(Y | X) P(X)}{P(Y | X) P(X) + P(Y | \neg X) P(\neg X)}$$

(5) **Application.**

$$\text{a. } P(\text{Eagle} | B) = \frac{P(B | \text{Eagle}) P(\text{Eagle})}{P(B | \text{Eagle}) P(\text{Eagle}) + P(B | \text{Non-eagle}) P(\text{Non-eagle})}$$

$$\begin{aligned} &= \frac{be}{be + 1 \cdot (1 - e)} \\ &= \frac{be}{be + (1 - e)} \left(= \frac{1}{1 + \frac{1 - e}{be}} \right). \end{aligned}$$

$$\begin{aligned} \text{b. } P(\text{Non-eagle} | B) &= 1 - \frac{be}{be + (1 - e)} \\ &= \frac{1 - e}{be + (1 - e)} \left(= \frac{1}{1 + \frac{be}{1 - e}} \right). \end{aligned}$$

Upon hearing an A-call, the receiver can infer that there is an eagle, since A-calls are only used for Eagles. When the receiver hears a B-call, the situation is more complex, as shown by (5). In all cases, after hearing a B-call, the hearer increases its assessment of the relative probability of a non-Eagle threat, compared to an Eagle-threat, as shown in (6).²

(6) $P(\text{Non-eagle} | B) > P(\text{Non-eagle})$

Proof: $be < e$ (since $0 \leq b < 1$), therefore $be + (1 - e) < e + (1 - e) = 1$, therefore $(1 - e)/(be + (1 - e)) > (1 - e)$. $\square$

To illustrate, in Table 2 the prior probability of a Non-eagle was very low (0.1), but the posterior probability of a Non-eagle given after hearing a B-call is, as mentioned above, approximately .18, which is still low, but nearly twice as much as the prior probability.

As a result, whatever the value of e and b , upon hearing a B-call, a Bayesian receiver should pay more attention than before to the possibility of non-eagle-related threats. This is true even in the absence of any Informativity Principle, and even if b is high, i.e. even if the B-call (i.e. the general call) tends to be favored across the board, in violation of the Informativity Principle. In particular, if eagle-related threats are no more frequent than non-eagle-related ones ($e \leq 0.5$), as we assumed in Table 1, then whatever the value of b is, $P(\text{Non-eagle} | B) > 0.5$.³ That is, in a situation where initially non-eagle-related threats and eagle-related threats are equally likely, B will lead to an inference that a non-eagle-related threat is likely as soon as the A-call is sometimes used for eagles, even if in general B-calls are preferred.

More generally, the formula in (5b) implies that the posterior probability of a non-eagle-related threat after hearing a B-call increases when e and b decrease. That is, the function

$$f(e, b) = \frac{1}{1 + \frac{be}{1 - e}} = P(\text{Non-eagle} | B)$$

²We are only talking here about the relative probabilities of the two types of threats *conditional on their being a threat*. When hearing a B-call, the receiver learns that there is a threat, so typically both eagle and non-eagle threats become more probable than before, and if eagle-related threats are antecedently much more likely than non-eagle-related ones, the receiver should still be more concerned about eagles than non-eagles after hearing a B-call, despite the fact that a B-call increases the probability share of non-eagle-related threats relative to that of eagle-related threats.

³If $e \leq 0.5$, $be/(1 - e) \leq 0.5 \times b/(1 - 0.5) = b$, hence $P(\text{Non-eagle} | B) = 1/(1 + be/(1 - e)) \geq 1/(1 + b) > 0.5$ (since $b < 1$).

is decreasing with respect to its two arguments.⁴ Intuitively, this is because when the prior probability of an eagle-related-threat (e) is low, its posterior probability will tend to be low as well, and if the probability ($= b$) of using a B-call in an eagle-related-situation is low, a B-call is even more indicative of a non-eagle-related threat.

Now in simple cases, we can assume that the receiver will adopt a non-eagle-related behavior just in case the probability of eagle presence after hearing B is less than a certain threshold θ , as stated in (7).

(7) The receiver will adopt a non-eagle-related behavior after hearing B just in case

$$P(\text{Eagle} \mid B) < \theta$$

i.e.

$$\frac{be}{be + (1 - e)} < \theta$$

i.e.

$$(1 - \theta) be < \theta(1 - e)$$

For concreteness, we can set $\theta = .5$, that is, the receiver adopts a non-eagle-related behavior if it's more likely that the threat is non-eagle- than eagle-related. The condition in (7) simplifies to (8).

(8) $be < 1 - e \iff e(1 + b) < 1$

We can see that if e is itself equal to or smaller than 0.5, this condition holds for all possible values of b (reason: since $(1 + b) < 2$, $0.5(1 + b) < 1$).

For simplicity, we have stated the predictions in terms of eagle-related vs. non-eagle-related threats, although analyses typically posit slightly different meanings, as will be seen in the empirical discussion below.

3 Why the simple Bayesian alternative doesn't derive key data

In their landmark study of call meaning in Vervet monkeys, Seyfarth et al. (1980) based their analyses on monkeys' reactions to playbacks of their own calls. These were thus comprehension data. If the animal Informativity Principle had been motivated by comprehension data as well, the foregoing remarks could suggest that the principle can be derived from Bayesian reasoning. But the empirical situation is entirely different: all the case studies that motivated the Informativity Principle were largely based on production data, as we will now see.

Titi monkeys Let us start with Titi monkeys. The analysis developed in Schlenker et al. (2016b) posits that B calls are general calls, to be used for any noteworthy event. By contrast, A-calls are used for diverse serious non-ground alerts (notably, eagle-related alerts). The key for our argument is that the main data come from production rather than comprehension experiments.

We have copied in Fig. 1 the results of Cäsar's experiment involving a model raptor in the canopy, with each line representing the first 30 calls produced. Virtually all of them are A-calls, there are a few O-calls or unidentified calls, and no B-calls at all.

By contrast, B-calls are used for predatory and non-predatory alerts alike (including for noteworthy events that are not dangerous, such as feeding or deer presence). This was taken by Schlenker et al. (2016b) (followed by Commier and Berthet 2019) to show that B-calls are general alert calls.

Interestingly, production data were complemented in this case with comprehension data. Cäsar (2012) assessed in detail Titi gaze orientation after hearing A- vs. B-calls. It is clear that receivers expected a non-ground threat after hearing A-calls, as such playbacks "elicited longer looking in the upwards direction, indicating monkeys were anticipating an important event, such as a raptor attack, from above" (Cäsar 2012 p. 165). For B-calls the situation is a bit less clear: "Playback elicited longer looking towards the caller,

⁴Proof: be is increasing in e , and so is $1/(1 - e)$, and therefore $eb/(1 - e)$ is increasing in e . $(be/(1 - e))$ is also of course increasing in b . So $[1 + be/(1 - e)]$ is increasing in both e and b , and therefore $1/[1 + be/(1 - e)]$ is decreasing in both e and b .

Leopard acoustic model

No.	N	1	2	3	4	5	6	7	8	9	10	11
1	14	P	P	P	P	P	P	P	P	P	P	P
2	4	P	P	P	P							
3	20	P	P	H	H	H	P	P	P	P	P	P
4	9	P	P	H	P	P	P	P	P	P		
5	5	P	H	H	P	H						
6	11	P	H	H	P	P	P	P	P	P	H	P
7	2	P	H									
8	3	H	H	P								
9	15	P	H	H	H	P	P	P	P	P	P	P
10	5	P	P	P	P	P						
11	14	P	P	P	P	P	P	P	P	P	P	P
12	3	P	P	H								
13	4	P	P	H	H							
14	4	P	H	H	H							
15	4	P	H	H	H							
16	2	P	P									
17												

Leopard visual model

N	1	2	3	4	5	6	7	8	9	10	11
7	P	P	P	P	P	P	P				
10	P	P	P	P	P	P	P	P	P	P	
37	P	P	P	P	P	P	P	P	P	P	P
63	P	P	P	H	H	P	P	P	P	P	P
8	P	H	H	P	P	P	P	P			
10	P	H	H	H	H	P	P	P	P	P	
3	P	H	H								
40	P	P	P	P	P	P	P	P	P	P	P
49	H	H	H	H	H	H	H	H	H	H	H
9	P	P	P	H	P	H	H	P	P		
6	P	P	P	P	P	P					
18	P	P	P	P	P	P	P	P	P	P	P
21	P	P	H	H	P	P	P	P	P	P	P
5	P	H	H	P	P						
10	P	P	P	P	P	P	P	P	P	P	P
16	P	P	P	P	P	P	P	P	H	P	P
9	P	P	P	P	P	P	P	P	P	P	P

Fig. 2a

Eagle acoustic model

No.	N	1	2	3	4	5	6	7	8	9	10	11
1	23	H	H	H	H	P	H	H	P	P	P	P
2	8	H	H	H	H	H	H	H				
3	29	H	H	H	H	H	P	H	H	H	H	H
4	13	H	H	H	H	H	P	P	P	P	P	P
5	38	H	H	H	H	H	H	H	H	H	H	H
6	33	H	H	P	H	P	H	H	H	H	H	H
7	12	H	H	H	H	H	H	H	H	H	H	H
8	12	P	P	P	P	P	P	P	P	P	P	P
9	40	H	H	H	H	H	H	H	H	H	H	H
10	13	H	H	H	H	H	H	H	H	H	H	H
11	20	H	H	H	H	H	H	H	H	H	H	H
12	5	H	H	H	H	H						
13	14	H	H	H	H	H	H	P	P	P	P	
14	19	H	H	H	H	H	H	H	P	P	P	
15	13	H	H	H	P	P	P	P	P	P	P	P
16	19	P	P	P	P	P	P	P	P	P	H	P
17	12	H	H	H	H	P	P	P	P	P	P	P
18	7	H	H	H	H	H	H					
19	2	H	H									
20	17	H	H	H	H	H	H	H	H	H	H	H

Eagle visual model

N	1	2	3	4	5	6	7	8	9	10	11
11	H	H	H	H	P	P	P	P	P	P	P
33	P	P	P	P	P	P	P	P	P	P	P
6	H	H	H	H	P	P					
8	H	H	H	H	P	P	P	P			
3	H	H	H								
57	H	H	H	H	H	H	H	H	H	H	H
5	H	H	H	H	H						
10	H	H	H	H	P	H	P	P	P	H	

Fig. 2b

Figure 2: Arnold et al.'s experiments with leopard growls and leopard visual models in (a), and eagle shrieks and eagle visual models in (b) (Arnold et al., 2008). Each line represents the series of the first eleven putty-nosed calls given in response to visual and acoustic models of a crowned eagle: hack calls (H) and pyow calls (P).

It should be noted that in some cases, eagle calls are followed by general calls. This pattern is illustrated in Putty-nosed sequences in Fig. 2b, and it is also found in Titi experiments (Cäsar 2012; Cäsar et al. 2013; Schlenker et al. 2016b). A standard line of analysis is that in some cases, the threat level typically decays after some time (Schlenker et al. 2016a,c). If Putty-nosed hack has a ‘serious non-ground’ component, alarm decay has the consequence that after a while hack can’t be used any longer (while there is a non-ground threat, it is a weak one), and thus only pyow can be used.

Blue monkeys Related conclusions were drawn by Schlenker et al. (2016c) on the basis of Blue monkey calls (unlike the other case studies discussed here, this proposal was part of a survey rather than of an article-length case study, and its conclusions are correspondingly more speculative). Blue monkeys have a common ancestor with Putty-nosed monkeys that lived 2.5 million years ago (Guschanski et al., 2013), and they have in part similar calls. In particular, they have pyow calls that occur in diverse environments, notably (but not exclusively) when there is a terrestrial predator, male-male agonism or agonism with another group (with presence in aerial predator contexts that is lower than expected by chance, but still significant). This is shown in Fig. 3 (Fuller, 2013).

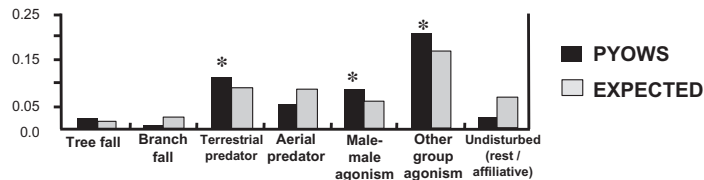


Figure 3: Fuller’s (2013) observational data with Blue monkey pyows.

From natural observations, comparison of observed to expected frequencies of pyows associated with each external variable. Y-axis shows proportion of all records. Pairs marked with * indicate significantly higher than expected association.

Schlenker et al. (2016c) postulated that, just as in Putty-nosed monkeys, pyows are general alert calls. But there are several other calls in the Blue monkey repertoire, for instance ka calls that are used (among others) for aerial predators. In order to explain why pyows do not occur in all environments, Schlenker et al. (2016c) once again appealed to the Informativity Principle. What matters for present purposes is that, whether right or wrong, this analysis was primarily based on production data.

Campbell’s monkeys Finally, we come to a more involved analysis, that of Campbell’s monkey calling behavior. This was the very first study that made use of the animal Informativity Principle, but we discuss it last because of its complexity (Schlenker et al., 2014, 2016d). The authors start from Campbell’s monkey calling behavior in the Tai forest (Ivory Coast), where the main predators are leopards and eagles. Campbell’s monkeys use krak calls to raise leopard alerts, and they use hok calls for raptor alerts. In addition, they have suffixed calls. Krak-oo is used for all sorts of alerts, and hok-oo for non-ground disturbances (Schlenker et al., 2016d).

Interestingly, calling behavior is in part different at another site, Tiwai island (Sierra Leone), where leopards haven’t been seen for dozens of years. There krak is used as a completely general call.

Schlenker et al. (2016d) summarize the proposed analysis as follows:

"As a first step, we take krak to trigger unspecific alerts and hok to trigger non-ground alerts. As a second step, to analyze the meaning of the suffix -oo, we assume that if R is krak or hok, R-oo indicates a weak alert of the R type. Thus, hok-oo indicates a weak (-oo) non-ground (hok) alert, which is more informative than hok.

It is in the third step that we make crucial use of the Informativity Principle: hok competes with other calls and because hok-oo (pertaining to weak non-ground alerts rather than to any non-ground alert) is more specific, the meaning of hok is enriched to hok but not hok-oo. Thus it applies only to aerial (hok) non-weak (not hok-oo) alerts – hence the raptor uses. By the same logic, the unspecific alert krak competes with krak-oo but also with hok. Due to this competition with two more-informative calls, ultimately krak can be

used only for serious (not *krak-oo*) ground (not *hok*) disturbances. This comes very close to the leopard uses observed in the Tai forest."

The fourth and final step is to account for the use of *krak* as a general call on Tiwai island. "Strikingly, this use corresponds to the basic (unenriched) meaning of *krak*. The question is why this bare meaning fails to be pragmatically enriched on Tiwai. A plausible answer is that this would yield a useless meaning due to the absence of serious ground predators."

The proposal, then, is that the use of *krak* on Tiwai island reflects its unadorned (and general) meaning. In the Tai forest, competition between *krak* and the more informative *krak-oo* (= non-serious alert) and *hok* (= non-ground alert) yields a strengthened meaning of "serious and ground alert".

Once again, our goal is not to adjudicate the debate regarding the analysis of Campbell's calls (Schlenker et al. 2014 only present the competition-based analysis as a serious contender, and certainly not as the only game in town). What is crucial for present purposes is that the key data come from production experiments in which calling behavior was triggered by predator sounds or naturalistic events. For instance, the data below were obtained from field experiments in which eagle shrieks and leopard growls were played in the Tai forest and on Tiwai island. They show that *krak* has a general use on Tiwai island but a leopard-related use in the Tai forest, as is expected given the initial generalizations.

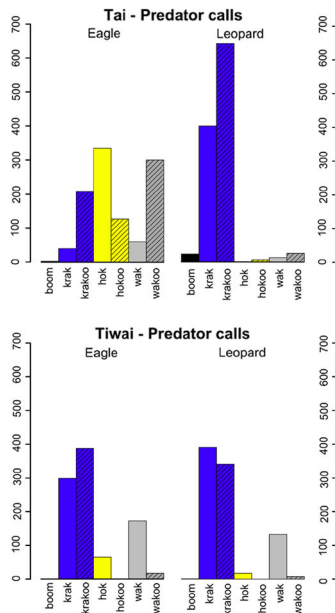


Figure 4: Number of calls of different types in response to different playback stimuli in Tai and on Tiwai

In sum, key data that motivated the Informativity Principle were based on production. The Informativity Principle in production captures these data. Simple Bayesianism does not: it is an inferential mechanism, which has nothing to say about the choice between two signals in production.

4 The division of labor between the Informativity Principle and Bayesian reasoning

We turn to the consequences of our remarks for animal semantics and its evolution.

Three Possibilities In view of the foregoing results, there are three conceivable ways of dividing the work between the Informativity Principle (viewed as primitive), pragmatics, and Bayesian reasoning.

1. In the human case, the Informativity Principle on the speaker's side is typically viewed as deriving from an ability to *reason* about both the listener's goals and the listener's interpretation of messages (Grice 1989

and much subsequent work). This derivation would make sense for our target data if there were evidence for a theory of mind in monkeys. But this is not currently the case.

In brief, there has been considerable work on the existence of a theory of mind in apes—and in particular on their ability to represent false beliefs. While the jury is still out, Krupenye and Call (2019) argued that some experiments have yielded a positive outcome. But they also noted that conclusions were different in monkeys, where results “raise the possibility that the mechanisms underlying apes’ sensitivity to others’ beliefs may have evolved uniquely in the ape lineage”.

Even without a theory of mind, however, one might imagine that the sender is able to observe the receiver’s *behavior*, and to reason based on this knowledge. In the future, a key question will thus be whether there is evidence for the *specific* cognitive abilities needed to derive the Informativity Principle in production from a reasoning on the receiver’s behavior. This is a far more fine-grained question than has been addressed so far. The literature has uncovered audience effects—instances in which a sender takes into account the receiver’s state of mind. A famous case involves chimpanzees that fail to produce snake alarm calls when their interlocutors are already aware of eagle presence (Crockford et al., 2012). In these cases, the sender might have a theory of mind, or it might observe that the receiver is better off hearing S than not hearing S , when S is true. But what we need to derive the Informativity Principle in production is something more sophisticated, namely the knowledge that the receiver is better off hearing S^+ rather than S when both are true.

A further difficulty bears mentioning. The cognitive abilities needed would have to be of a rather abstract sort. In Putty-nosed monkeys, Blue monkeys and Campbell’s monkeys, alarm calls discussed above are produced by a resident male for an entire group it may not see (as the group is spread in trees). For this reason, the resident male would have to reason about the behavior of a very abstract addressee.

Even without this constraint, it is hard to see how the sender could easily infer how messages affect the receiver’s behavior (note that, by contrast, the sender’s behavior is trivial to observe). With the constraint that resident males often don’t see the receivers, the problem becomes even harder, short of representing the receiver’s cognitive behavior.

We conclude that, in view of the available evidence, it is not plausible to derive the Informativity Principle in production from a reasoning on the receiver’s behavior.

2. A plausible alternative is that the Informativity Principle (taken as primitive) and Bayesian reasoning play complementary roles: the Informativity Principle is responsible for patterns of production, while in comprehension it is not primitive but follows from Bayesian reasoning.

Technically, the reason this result can be derived is that $b = P(B \mid \text{Eagle}) = 0$, which essentially guarantees that the inequality in (7) is satisfied as soon as there are non-eagle situations (i.e. $e \neq 1$).⁵

In the case of human language, the same result can be conceptualized in two slightly different ways: (i) the addressee has access to the speaker’s strategy, and thus infers from the production of a less informative expression that the more informative one was false – as in the standard Gricean approach; (ii) the addressee has access to the speaker’s behavior.⁶ In the animal case, (i) might make little sense if no theory of mind is present, but (ii) makes good sense.

The proposed synthesis represents a division of labor that is not usually considered in humans (where reasoning can do everything), nor in prior work on animal linguistics. For fear of unduly committing to a theory of mind in animals, earlier animal literature took the Informativity Principle to be primitive in both its production and its comprehension sides. As we have now seen, for the comprehension side, this measure was unnecessary.

Importantly, on this view the Informativity Principle in production could be probabilistic rather than deterministic. In our stylized example above, for the Informativity Principle in comprehension to be derived, it is not necessary that $b = P(B \mid \text{Eagle}) = 0$, as long as the inequality in (7) is satisfied. This could be empirically important. One argument adduced by Schlenker et al. (2014) to treat Campbell’s monkey *krak* as

⁵In fact, given that $P(B \mid \text{Eagle}) = 0$, Bayesian reasoning boils down here to realizing that since B is never used for an Eagle-stimulus, it necessarily indicates a non-Eagle-stimulus.

⁶In both the neo-Gricean tradition (Grice 1989; Horn 1972; Gazdar 1979; Hirschberg 1991; Sauerland 2004; Schulz and van Rooij 2006; Spector 2007, a.o.) and in the Bayesian pragmatics literature (Frank and Goodman 2012; Goodman and Stuhlmüller 2013; Franke and Jäger 2016; Degen 2023, a.o.), the pragmatic listener draws inferences on the basis of their knowledge of the speaker’s behavior in different situations. This knowledge can in principle come either from reasoning about what a rational speaker would do, or from *observing* what the speaker typically does. While the first option (reasoning about the speaker’s optimal strategy) is typically assumed in human pragmatics, it is not a necessary component of the analysis.

a general call is that even in the Tai forest, where it has been described as a leopard-related call, it appears in a small number of cases in eagle-related environments (and the authors point out that this "overextension" of *krak* is greater than the overextension of the eagle call, *hok*, to leopard-related situations; this suggests that mere production errors are not responsible for the overextension of *krak*; rather, the core meaning of *krak* might be general). If so, the Informativity Principle in production is not invariably applied. Still, the exceptions might be rare enough that the Bayesian derivation of the Informativity Principle in comprehension still goes through.

3. A more conservative view is, of course, that the Informativity Principle is primitive both in its production and in its comprehension sides. On the assumption that Bayesian reasoning is prevalent in animals, as discussed above, one would need to justify why the comprehension side is needed.

In sum, the most appealing possibility is arguably that the Informativity Principle is primitive in production, and derived from Bayesian reasoning in comprehension.

Evolutionary consequences In recent work, it was argued that simple notions of evolutionary stability in the tradition of Maynard Smith can explain the emergence of the Informativity Principle, viewed (as in all earlier animal linguistics) as primitive (Schlenker et al., forthcoming). It was proposed that there is a clear evolutionary path in which the principle appears in production first, and in comprehension second. The reason was that this evolutionary sequence ensures (irrespective of the detailed pay-offs) that each step increases shared utility and doesn't create misunderstandings. By contrast, some misunderstandings would emerge if the Informativity Principle were applied in comprehension before it arose in production. To see this, suppose a Titi monkey produces a B-call (general alert) in an eagle-related situation, a behavior that is possible without the Informativity Principle in production. A receiver who applied the Informativity Principle in comprehension would wrongly infer that this was not an eagle-related situation, where the A-call would be expected. Miscommunication would thus ensue.

This reasoning is valid if two evolutionary steps are in fact needed. But our analysis suggests that the second step might be unnecessary—accounting for the emergence of the Informativity Principle in production might be all that is needed if Bayesian reasoning automatically fills in the second step. This could simplify the analysis of the evolutionary emergence of the Informativity Principle.

In addition, future work might have to revisit the details of the first step. In a nutshell, on the assumption that the receivers perform Bayesian reasoning on the sender's behavior, a mutant that adopts the Informativity Principle in production might produce more shared utility than was envisioned in earlier models, which took the receiver *not* to be Bayesian.

5 Conclusion

In sum, while simple Bayesianism could in principle obviate the need for some uses of the Informativity Principle applied in comprehension, this is not so for the Informativity Principle applied in production. Key data that motivated the animal Informativity Principle were based on production, not comprehension. Thus either senders can reason on the receivers' Bayesian behavior, or the Informativity Principle must be taken as primitive in production. To the extent that the first disjunct is implausible (in view of current data on monkeys' theory of mind), the second disjunct must be accepted. This makes plausible a new division of labor between the Informativity Principle and Bayesian reasoning: the principle must be taken as primitive in its production side but it can be taken as derived in its comprehension side. This has consequences for the evolution of the Informativity Principle. Recent studies took the principle to arise from a two-step evolutionary process—plausibly with production first, and comprehension second. Our analysis shows that the second step could be dispensed with.

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